

From two records to a match probability

Constructed records; stipulated probabilities.

Record a in source A

Name: Amina Patel
Birth date: 1988-04-12
Postcode: 3000

Record b in source B

Name: Amina Patel
Birth date: 1988-04-12
Postcode: 3000

Compare the fields in this order: name, birth date, postcode

$$\gamma(a, b) = (\text{agree}, \text{agree}, \text{agree})$$

Look up each outcome rate among matches (m_j) and nonmatches (u_j)

The field weight is $w_j = \log_2(m_j/u_j)$: A positive weight supports a match.

Field	Outcome	m_j	u_j	m_j/u_j	Weight (bits)
Name	agree	85%	1%	85	+6.4
Birth date	agree	90%	0.2%	450	+8.8
Postcode	agree	60%	3%	20	+4.3

Start with prior $\pi = 0.01\%$

1 match in 10,000 pairs
prior log odds: -13.3 bits

Add prior and field weights

Prior + name + birth
date + postcode

$$W \approx 6.3 \text{ bits}$$

Estimated match
probability

98.7%

$$P(M \mid \gamma) = \frac{2^W}{1 + 2^W}$$

Change only record b 's birth date to 1975-11-03

$\gamma(a, b) = (\text{agree}, \text{contradicts}, \text{agree})$.

Birth-date evidence: $m_j = 2\%$, $u_j = 90\%$, so $w_j \approx -5.5$ bits.

New score: $W \approx -8.0$ bits. **Estimated match probability: 0.4%.**

This model assumes conditional independence within matches and within nonmatches.

Totals use unrounded weights. A decision also needs a chosen threshold.